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PALEOMAGNETIC POLES AND POLARITY ZONATION FROM CAMBRIAN AND DEVONIAN STRATA OF ARIZONA

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Basal Paleozoic Tapeats Sandstone (Early and Middle Cambrian) in northern and central Arizona exhibits mixed polarity and a low-latitude paleomagnetic pole. Carbonates of Middle and early Late Cambrian age, and directly superposed carbonate and carbonate-cemented strata of latest Middle(?) and early Late Devonian age, are characterized by reversed polarity and high-latitude poles. The high-latitude Middle Cambrian pole, which appears to record a large but brief excursion of the polar wandering path, is considered provisional pending additional work. The Devonian data from Arizona indicate that a shift of the pole to a "late Paleozoic" position had occurred by Middle Devonian time.

1. Introduction

The work reported here is part of a larger study of the magnetostratigraphy of Precambrian and Paleozoic strata in the western United States. Investigations in Phanerozoic rocks have the goal of relating polarity zonations and poles to fossiliferous rocks in order to establish time lines and thus to obtain new information for the dating of unfossiliferous strata. Because stratigraphic horizons at which changes in polarity and shifts in pole position occur need to be identified, oriented core samples were obtained on close stratigraphic spacings across the sampled sections.

2. Stratigraphy

The strata investigated crop out in central Arizona and in the central and eastern Grand Canyon of northern Arizona (Fig. 1). In northern Arizona, units sampled (Fig. 2) include the Tapeats Sandstone (Early and Middle Cambrian), the Bright Angel Shale and Muav Limestone (Middle Cambrian), and the overlying Temple Butte Limestone (latest Middle? and early Late Devonian on the basis of conodonts identified by Dietmar Schumacher, written communication, 1976). In central and southeastern Arizona, units sampled include the Tapeats Sandstone (Early and Middle Cambrian), the Abrigo Formation (Middle and Late Cambrian), and the Beckers Butte and Jerome Members (latest Middle? and early Late Devonian) of the Martin Formation. In central Arizona, the Beckers Butte rests directly on the Tapeats, which is preserved in topographically low areas on the Precambrian erosion surface (Fig. 2).

Paleomagnetic results are reported for hematitic or red-bed strata in the sections. Results reported for the Tapeats Sandstone and Temple Butte Limestone of Grand Canyon, and for the Beckers Butte Member of the Martin Formation of central Arizona, represent data collected at close stratigraphic spacings across the entire formations. Results reported for the Muav Limestone of the Grand Canyon, the Abrigo Formation of southeastern Arizona, and the Jerome Member of the Martin Formation of central Arizona represent data collected from discrete red-bed horizons within the formations. The stratigraphic intervals for which paleomagnetic data are reported and the polarity of the sampled intervals are shown in Fig. 2.

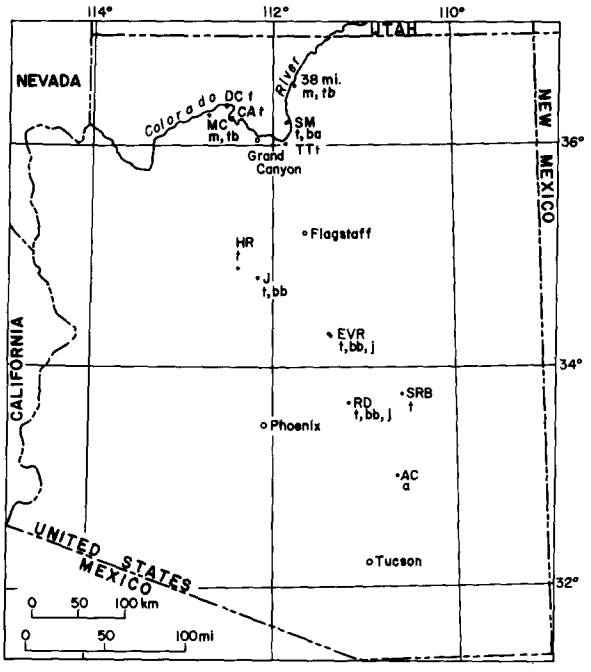


Fig. 1. Index map of Arizona. MC = Matkatamiba Canyon; DC = Deer Creek; CA = Conquistador Aisle; SM = Sixtymile Canyon; TT = Tanner Trail; 38 mi = mile 38 on Colorado River; AC = Ash Creek; SRB = Salt River Bridge; RD = Roosevelt Dam; EVR = East Verde River; J = Jerome; HR = Hubbel Ranch. Rock units: tb = Temple Butte Limestone; m = Muav Limestone; ba = Bright Angel Shale; t = Tapeats Sandstone; j = Jerome Member of Martin Formation; bb = Beckers Butte Member of Martin Formation; a = Abrigo Formation.

Collinson and Runcorn [1] sampled the Tapeats Sandstone and Bright Angel Shale south of the western Grand Canyon. Mean directions for these units were not determined because thermal cleaning had not been carried out to remove secondary magnetization acquired in the present-day field.

3. Field and laboratory methods

Oriented cores 2.5 cm in diameter and 5–10 cm long were obtained by means of a portable diamond core drill. Orientations were measured with compass and clinometer, read to the nearest one-half degree. Stratigrapic separations commonly were about 1/3 m or less, and in most cases one core was collected at each horizon.

After trimming, the natural remanent magnetization (NRM) of the cores was measured either on an air-driven spinner magnetometer having a background noise of 2×10^{-7} emu, or on a cryogenic superconducting rock magnetometer having a background noise of 1×10^{-7} emu or less. Pilot samples were subjected to progressive thermal demagnetization at six steps from 200° to 590°C (Fig. 3) for the determination of intervals of stable or characteristic magnetization. Thermal demagnetization analysis was employed because the remanent magnetization resides principally in hematite, whose magnetic coercivity is very high. During the early stages of investigation of the Tapeats Sandstone and the Temple Butte Limestone of the Grand Canyon, several pilot samples also were subjected to progressive stepwise alternating field demagnetization. Peak fields of up to 5000 Oe generally resulted in less complete cleaning in comparison to results obtained from thermal demagnetization. As shown in Fig. 3, a substantial part of the NRM remains after thermal cleaning up to 590°C, indicating that much of the stable remanence resides in hematite rather than magnetite, which has a Curie temperature of 580°C.

4. Paleomagnetic directions

NRM and thermally cleaned directions for the Cambrian and Devonian formations are summarized in Table 1. Equal area plots of the cleaned directions (declinations and inclinations) for individual samples are shown in Fig. 4, as well as mean NRM and mean cleaned directions. The data reveal that, for the samples shown on the plots, components of the present field directions acquired during recent weathering were removed by the thermal demagnetization. The Tapeats Sandstone, and very locally parts of the Temple Butte Limestone in the eastern Grand Canyon, are fairly porous and in places show the effects of bleaching or alteration by groundwater. A number of samples (not reported here) displayed magnetic directions that cluster around the present field direction implying that secondary chemical remagnetization occurred in the geologically recent past during erosion and exposure of the beds. Thermal demagnetization successfully removed this late component of magnetization for some but not all of the samples. In contrast,

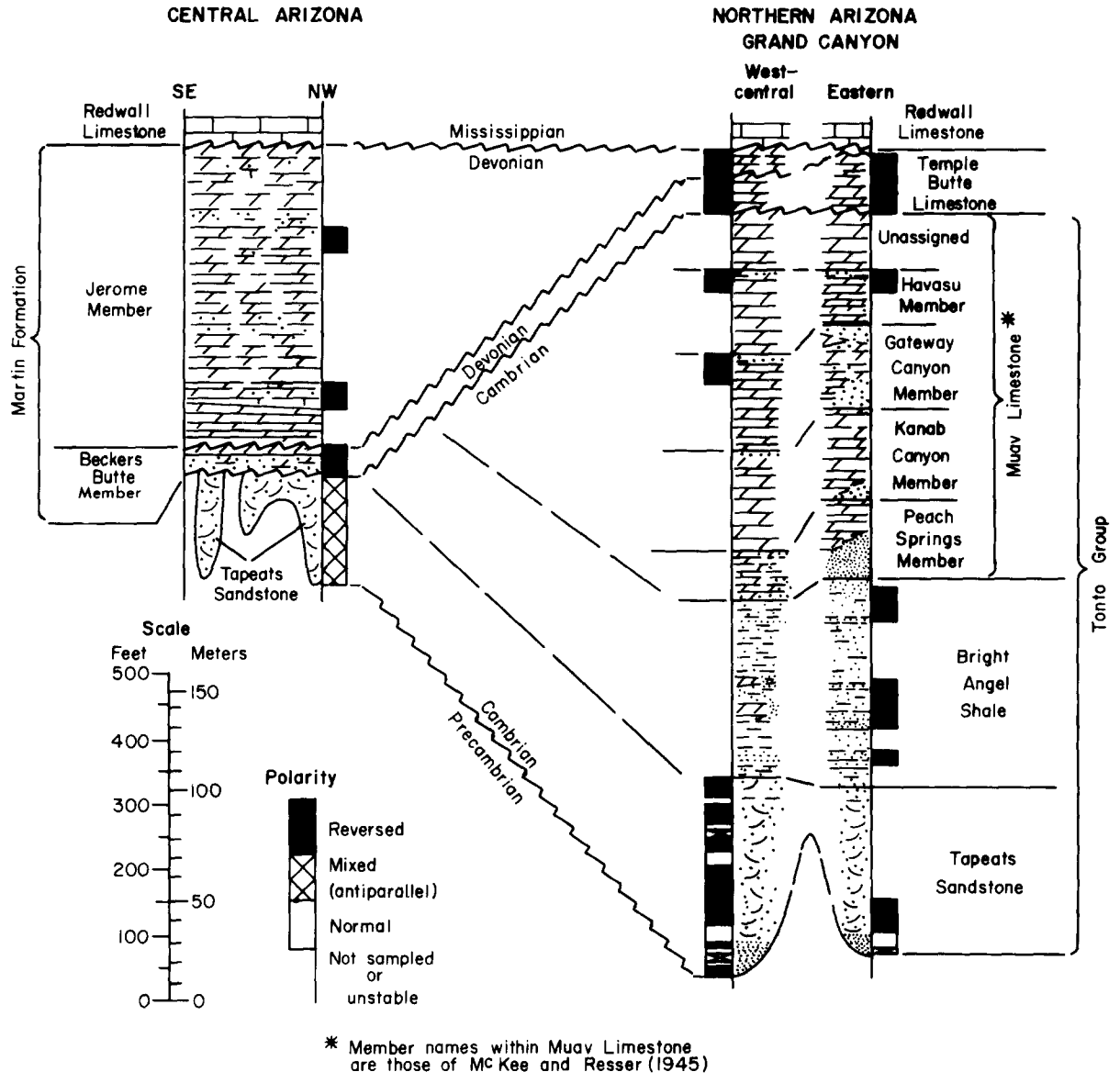


Fig. 2. Generalized sections of Cambrian and Devonian strata in northern and central Arizona, showing polarity zonation.

the NRM directions for samples of the Cambrian Muav Limestone and for many samples of the Devonian rocks of central and northern Arizona are well clustered and are far from the present field direction. Only small changes in direction occurred with their demagnetization, attesting to the virtual absence of recently acquired secondary components of magnetization (Fig. 4).

Stable characteristic directions were determined by the endpoints of demagnetization paths that lie farthest from the present field direction. These directions for the most part also show the best groupings (minimum dispersion). Demagnetization steps used for the calculation of characteristic directions for each unit are listed in Table 1.

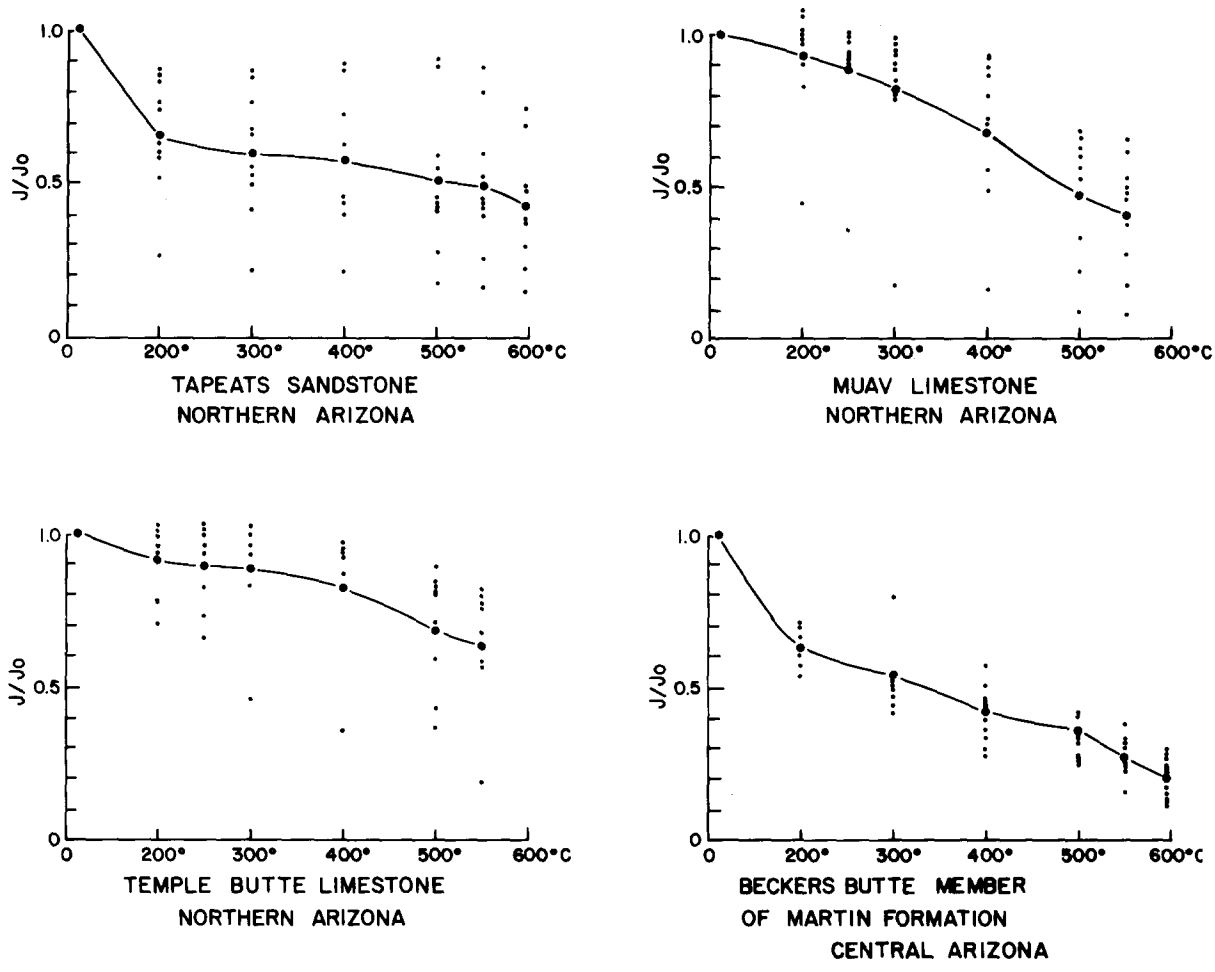


Fig. 3. Plots of normalized magnetization intensities with progressive thermal demagnetization. Large dots are means for the samples.

Tapeats Sandstone. The Tapeats Sandstone, of Early and Middle Cambrian age [2], was sampled principally at two localities in the Grand Canyon – basal beds exposed at the head of the north fork of Sixtymile Canyon in the eastern Grand Canyon, and a complete section exposed along the Colorado River 1–2 km west of Deer Creek in the west-central Canyon, where nearly the entire Tapeats is red and largely unbleached. Several additional samples were obtained from the basal Tapeats west of the Tanner Trail and in Conquistador Aisle.

The directions of magnetization for the Tapeats in the Grand Canyon (Fig. 4) are reported here only for those samples that exhibited stable remanent directions apparently free from contamination by the present field after thermal cleaning. Stable endpoints were arrived at in the range of 500°–590°C. Eighty-five of 166 samples were excluded from the Deer Creek section for the calculation of a pole, and nine of 42 samples were excluded from the Sixtymile Canyon section. The discarded samples exhibit scattered directions or directions "streaked" toward the present axial field direction. More than 12 polarity reversals were found in the lower and middle parts of the Tapeats in the Deer Creek section. In contrast, the upper part of the Tapeats is dominantly if not entirely of reversed polarity (Fig. 2). As shown in Fig. 4 and Table 1, the normal and reverse polarity

TABLE 1

Age and stratigraphic unit	Treatment [1]	Polarity [2]	Sites	N	Vector sum R	Declination (degrees)	Inclination (degrees)	K [3]	α_{95} [4]	Pole		δ_m [5]	δ_m [5]
										lat.	long.		
Devonian													
[Late Middle(?) and early Late]													
Temple Butte Limestone, Grand Canyon	NRM	R	2	80	73.748	149.7	− 7.4	12.6	4.6	47.0°S	64.7°W	4.7	2.4
	500°C	R	2	76	75.093	153.2	−15.7	82.7	1.8	52.6°S	65.1°W	1.8	0.9
Beckers Butte and Jerome Members of Martin Formation, central Arizona	NRM	R	3	66	60.408	152.6	− 4.0	11.6	5.4	49.0°S	67.1°C	5.4	2.7
	500°C	R	3	66	65.018	158.4	−12.7	66.2	2.2	55.7°S	70.8°W	2.2	1.1
Cambrian													
[Middle and Late]													
Abrigo Formation, southeastern Arizona	NRM	R	1	10	9.959	169.4	− 3.5	222.3	3.2	57.2°S	90.7°W	3.3	1.6
	500°C	R	1	10	9.970	168.8	− 6.5	301.5	2.8	58.5°S	91.1°W	2.8	1.4
[Middle]													
Muav Limestone, Grand Canyon	NRM	R	2	56	47.730	152.0	− 6.4	6.7	8.0	48.0°S	68.3°W	8.0	4.0
	500°C	R	2	56	54.667	157.0	−16.3	41.3	3.0	55.0°S	70.4°W	3.1	1.6
[Early and Middle]													
Tapeats Sandstone, Grand Canyon	NRM	N	3	62	51.842	292.0	38.3	6.0	8.1	29.9°N	163.9°E	9.5	5.6
	500–590°C	N	3	62	58.068	275.1	8.0	15.5	4.7	6.5°N	158.0°E	4.8	2.4
	NRM	R	3	67	58.757	82.3	16.1	8.0	6.5	11.0°N	24.4°W	6.7	3.5
	500–590°C	R	3	67	62.437	93.1	− 3.6	14.5	4.7	3.6°S	22.6°W	4.7	2.4
Combined N and R	500–590°C	M	4	129	120.400	274.0	5.7	14.9	3.3	4.9°N	157.7°E	3.3	1.7

[1] NRM = natural remanent magnetization; temperature(s) indicate thermal step(s) found to produce the best groups lying farthest from the direction of the present field.

[2] N = normal; R = reversed; M = mixed.

[3] $K \approx (N - 1)/(N - R)$.

[4] Semi-angle of the cone of 95% confidence level.

[5] Semi-major and semi-minor axes of the oval of 95% confidence level.

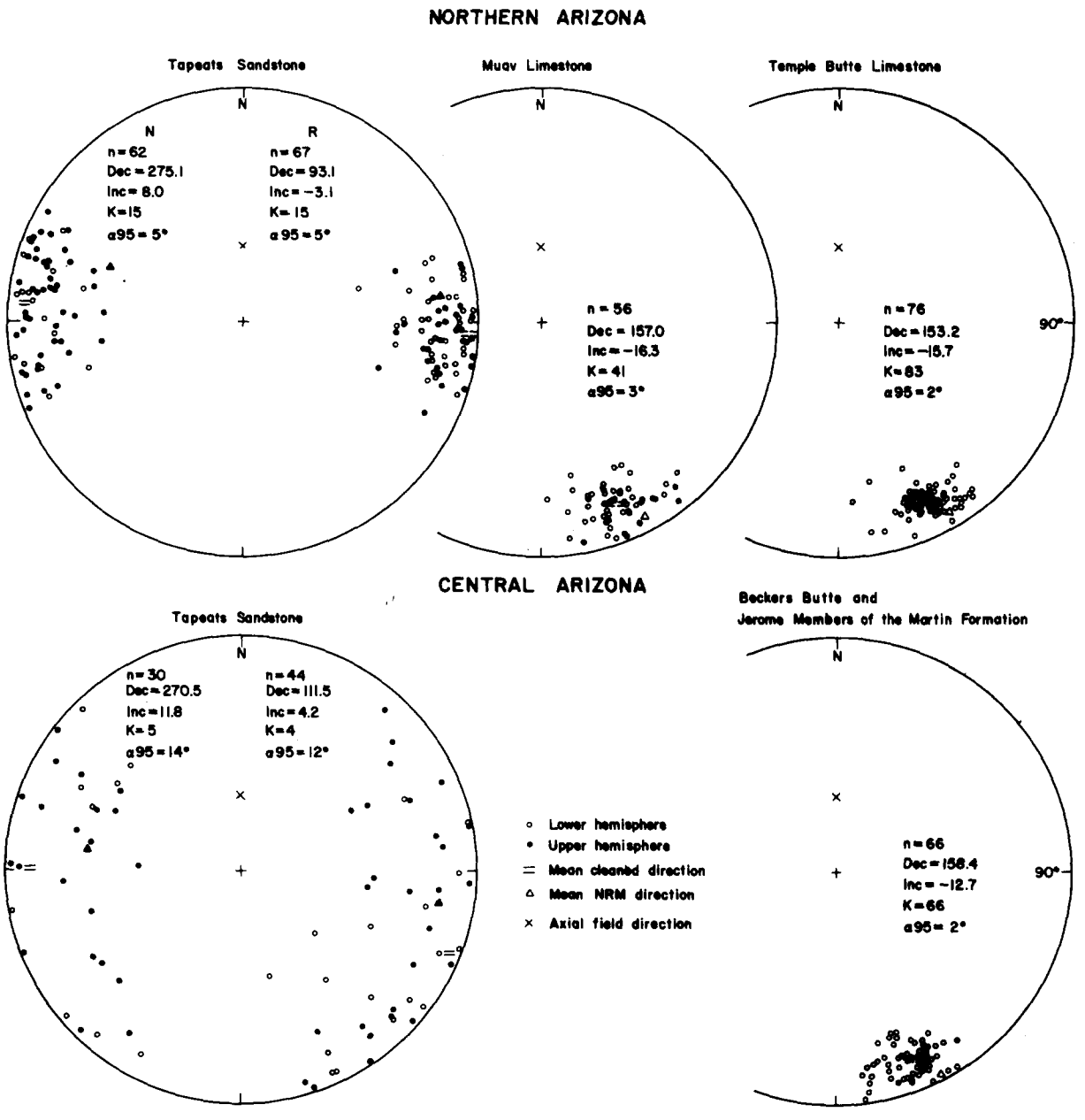


Fig. 4. Stereographic plots of paleomagnetic directions for selected Cambrian and Devonian rocks of northern and central Arizona.

directions from the selected samples are closely antiparallel. This is strong evidence for penecontemporaneous acquisition of magnetization during deposition of the Tapeats and also is evidence that present field components of magnetization were successfully removed from these samples by the cleaning. The closeness to antipolarity allows a mean Tapeats direction to be calculated by combining the normal poles and antipoles of the reversed polarity directions (Table 1).

The Tapeats Sandstone of central Arizona, sampled at four localities, also exhibits both normal and reversed polarity (Fig. 4). The paleomagnetic direction and mixed polarity have allowed the unfossiliferous basal Paleozoic sandstone of central Arizona to be correlated with and to be identified as Tapeats Sandstone. However, the scatter in directions is much greater than that seen in the Tapeats of the Grand Canyon of northern Arizona. This can be explained by the presence of two components of secondary magnetization in the population of data points. Data points that form a "streak" toward the present field direction (and many such points exist but have not been shown in Fig. 4) reflect recent remagnetization such as seen in the Tapeats of the Grand Canyon. However, the Tapeats in central Arizona is preserved in incomplete sections that are directly and unconformably overlain by Devonian strata (Fig. 2). Locally, bleaching of the uppermost Tapeats can be demonstrated on geologic grounds to have occurred before deposition of the Devonian strata. At one place the coloration of the overlying Devonian strata, and a Devonian paleomagnetic direction is superimposed at the top of the bleached zone. Thus, unlike northern Arizona where the Tapeats was protected from post-Cambrian near-surface alteration by more than 200 m of overlying Cambrian strata, the Tapeats of central Arizona lay exposed at the surface for tens of millions of years, if not more than 100 m.y., before depositon of the Devonian strata. Data points that form "streaks" from both the normal and reversed Tapeats directions toward the Devonian reversed direction reflect partial to locally nearly complete Devonian or pre-Devonian remagnetization of the central Arizona Tapeats. The mean directions shown in Fig. 4 for the Tapeats in central Arizona thus reflect both an original magnetization of Cambrian age and a secondary remagnetization of Devonian or pre-Devonian age.

Bright Angel Shale. The Bright Angel Shale, of Middle Cambrian age in the central and eastern Canyon [2], is a fairly impermeable siltstone and sandstone unit that overlies the Tapeats Sandstone in the Grand Canyon of northern Arizona (Fig. 2). The Bright Angel has been sampled for paleomagnetic analysis, but work has not yet been completed. Preliminary results from a few red intervals in the section suggest that the Bright Angel is entirely of reversed polarity, similar to the upper part of the Tapeats, and that its paleomagnetic direction lies near that of the Tapeats Sandstone.

Muav Limestone. The Muav Limestone, of Middle Cambrian age [2], has been sampled at two localities in the Grand Canyon (mile 38 on the Colorado River, and in Matkatamiba Canyon in the west-central part of the Canyon). Samples were obtained from hematite-laminated carbonate rock, and from two horizons of red sandy carbonate and calcareous sandstone deposited at times of regression of the sea. The last two are stratigraphic intervals that mark member boundaries in the Muav Limestone (Fig. 2). The red units give every appearance in the field of being fresh and unmodified from the time of deposition, and they lie 20–70+ m beneath Devonian strata that disconformably overlie the Muav. In the eastern Grand Canyon alteration that reflects deep weathering before deposition of Devonian beds has affected the upper 20–30 m of the Muav. The alteration is greenish yellow and limonitic, and is in no way similar to the hematitic material sampled from fresh and apparently unmodifed strata well within the Muav, or from overlying Devonian strata. Because the weathered zone lies between and separates apparently unmodified Cambrian and Devonian strata that contain stable directions of magnetization, there is no field evidence for invoking post-Muav alteration and remagnetization to account for the high-latitude, reversed polarity direction in the Muav, a direction that is similar to the direction obtained from Devonian strata (Table 1 and Fig. 4).

Abrigo Formation. Support for the Cambrian age of the Muav paleomagnetic direction is seen in a direction derived from a 1-m-thick hematitic oolite bed in the Abrigo Formation (Middle and Late Cambrian) sampled in Ash Creek, about 13 km east of Winkleman in southeast Arizona (Table 1 and Fig. 1). The oolite bed lies about 26 m below the top of the Abrigo and is overlain and underlain by rock containing inarticulate brachiopods considered by Dietmar Schumacher (oral communication, 1976) to be either late Middle or early Late Cambrian in age. The Abrigo paleomagnetic direction is similar to the Muav direction, with the small difference probably reflecting

local structural rotation on the monocline that contains the Abrigo outcrop. Thus, if the Muav and Abrigo directions are accepted at face value, a large and abrupt shift in paleomagnetic direction appears to have occurred in Middle Cambrian time.

Devonian strata. Samples from five sections of Devonian strata sampled in northern and central Arizona provide tightly grouped directions of reversed polarity (Fig. 4). As was the case for the Cambrian Muav Limestone and Abrigo Formation, the NRM directions lie close to the thermally cleaned direction (Table 1), revealing only a subordinate secondary magnetization in the NRM that can be attributed to the present-day field. The age of the Devonian beds for which paleomagnetic directions are reported is late Middle(?) Devonian (latest Givetian?) and early Late Devonian (Frasnian), on the basis of conodonts from the Temple Butte Limestone of the Grand Canyon (Dietmar Schumacher, written communication, 1976), and the age of psilophyte flora [3] and macrofossils [4] in the Devonian strata of central Arizona.

5. Discussion

With the exclusion of samples that display components of recent and Devonian remagnetizations, the Arizona Cambrian and Devonian paleomagnetic directions appear to have been acquired during or shortly after deposition of the beds. The existence of numerous polarity reversals in the Tapeats Sandstone is strong evidence for penecontemporaneous acquisition of the remanence and evidence against a general pervasive remagnetization of the beds. Remagnetization of the Tapeats in the present-day field direction is clearly recognizable in both northern and central Arizona, and in a number of cases this secondary magnetization was successfully removed by thermal cleaning. Additionally, a Devonian remagnetization of parts of the Tapeats is recognized in central Arizona, where Devonian beds were deposited unconformably on a bleached surface of the Tapeats. In spite of locally extensive Devonian and recent remagnetization of parts of the Tapeats in central Arizona, some antiparallel paleomagnetic directions have been obtained.

In contrast to the fairly scattered paleomagnetic directions seen in samples from the Tapeats Sandstone, samples from the Middle Cambrian Muav Limestone and from the Devonian strata of northern and central Arizona are unusually tightly clustered, and the NRM directions moved little with thermal cleaning. The directions from these strata are characteristic of a "late Paleozoic" direction. However, there is no evidence of "streaking" that would suggest post-Devonian (late Paleozoic) remagnetization of these beds. In light of the Tapeats paleomagnetic data, remagnetization is an imperfect process and some vestige of an earlier magnetization should be seen if the Middle Cambrian and Devonian strata of Arizona had been subjected to a late Paleozoic remagnetization. No such vestige exists.

Additionally, the well clustered directions seen in the Devonian in central Arizona are restricted to the comparatively impermeable Devonian beds. If these directions are the result of a pervasive late Paleozoic remagnetization then the underlying permeable Tapeats Sandstone should also have undergone pervasive remagnetization. The largely unmodified bleached zone of the Tapeats that underlies the unconformity is evidence against a pervasive Devonian or later remagnetization, evidence that is supported by mixed polarity directions in sandstone of the Tapeats. Lastly, field relations in the eastern Grand Canyon do not support a remagnetization hypothesis for the Muav Limestone, but rather indicate that hematite in the Muav is older than the development of a limonitic weathered zone at the top of the Muav. The greenish limonitic alteration can be seen to cross-cut hematitic carbonate rocks along joints near the base of the altered zone, which is evidence for the older age of the hematite. Field relations thus support the idea that hematite in the Cambrian and Devonian strata is not the product of either a late Paleozoic or a Devonian alteration and remagnetization.

A curious correlation does exist, however, between lithology and paleomagnetic direction. The lower siliceous and argillaceous clastic rocks yield a "Cambrian" direction, whereas overlying carbonate-cemented sandstone and carbonates yield a "Devonian" direction. One possible (although highly speculative) situation might have resulted in the imposition of a Devonian direction on the Cambrian

carbonate strata that lay near the regional surface of non-deposition and erosion. If *entirely* goethite rather than hematite had been deposited during the regressive stages of deposition of the members of the Muav Limestone and during deposition of the oolite bed in the Abrigo Formation, and if these strata were not deeply buried and remained saturated with ground water from Middle Cambrian until Devonian time, then a dehydration of the goethite accompanying exposure of the upper parts of the Muav and Abrigo in Devonian time could have led to the formation of hematite and the acquisition of a Devonian paleomagnetic direction. (For discussions of some of the properties of goethite and hematite, see Strangway et al. [5] and Larson and Walker [6]). Such an alteration history might not be evident at the outcrop and could account for the very well-clustered "Devonian" paleomagnetic directions seen in the Muav and Abrigo. However, the existence of any early hematite would result in paleomagnetic directions streaked toward the Tapeats direction, and such streaking is not observed. That all the iron oxides were initially hydrous and that they remained in this condition for about 200 m.y. seem geologically unlikely, although such a history cannot be unequivocally discounted at this time. Because of this, paleomagnetic directions reported for the Muav and Abrigo, although supported by field geologic relationships, are considered provisional pending additional work, particularly the acquisition of directions obtained from strata of similar age in which the remanence resides in magnetite rather than hematite. Note should be made that the dehydration of the assumed goethite could have taken place any time after deposition of the host beds and, if during the Cambrian, would reflect a Cambrian rather than a Devonian remagnetization.

6. Poles

North poles for the Tapeats Sandstone (Early and Middle Cambrian), Muav Limestone (Middle Cambrian), Abrigo Formation (Middle and Late Cambrian), Temple Butte Limestone (late Middle? and early Late Devonian), and Martin Formation (late Middle? and early Late Devonian are shown in Fig. 5. The pole for the Tapeats lies in the vicinity of Cambrian poles for North America reported elsewhere

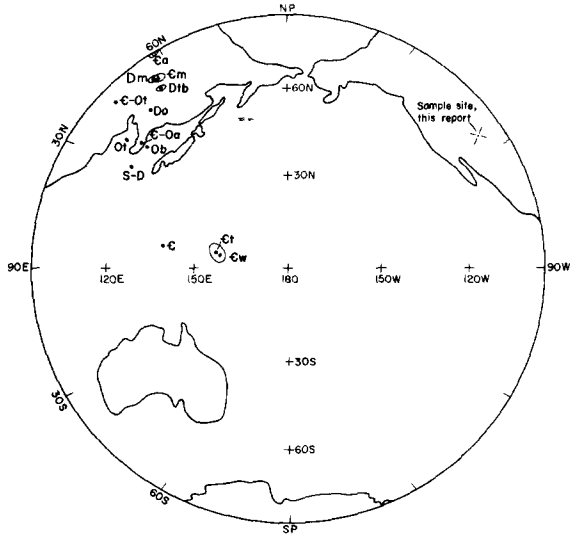


Fig. 5. North poles for Cambrian and Devonian strata of Arizona, and Paleozoic poles reported from elsewhere in North America. *Arizona poles*: Dm = Martin Formation; Dtb = Temple Butte Limestone; €a = Abrigo Formation; €m = Muav Limestone; €t = Tapeats Sandstone (ovals drawn at 95% confidence level). *Other North American poles*: Do = Middle Devonian limestone in central Ohio [15]; Ob = Late Ordovician Beemerville Alkaline Complex, New Jersey [17]; Ot = Middle Ordovician Trenton Limestone [16]; €-Oa, €-Ot = poles from plutonic rocks of Cambrian to Ordovician age from Colorado determined by thermal (t) and alternating field (a) demagnetization analysis [10]; S-D = mean Silurian and Devonian pole for North America [8]; € = mean Cambrian pole for North America [8]; €w = Late Cambrian Wilberns Formation [14].

[7; 8, p. 202]. The path for North American apparent polar wandering during the Paleozoic has been portrayed as proceeding in a rather regular manner from a low-latitude position west of 180° longitude during the Cambrian, northwesterly across the Pacific and into Siberia by Permian time [8, p. 202, fig. 110, p. 254, fig. 134]. Poles for the Cambrian Muav and Abrigo and for the Devonian strata of Arizona are far north of their expected positions, and they lie near the mean Lower Triassic pole reported by McElhinny [8].

The high-latitude Middle Cambrian pole in Arizona, on the basis of field geologic relations discussed above, may reflect the direction of the magnetic field at the time of deposition of the beds. This pole, however, is considered provisional pending additional work. High-latitude poles also have been ob-

seved in other Cambrian units of North America and include the Middle Cambrian Rome Formation of the central Appalachians [9], Cambrian intrusive rocks in Colorado (R.B. French, D.H. Alexander and R. Van der Vöo, written communication, 1976), and the same intrusives investigated by Larson and Mutschler [10], granites from the Wichita Mountains, Oklahoma [11], and the Lamotte Formation of Missouri [12] investigated by Al-Khafaji and Vincenz [12]. The high-latitude pole has been attributed to a Late Paleozoic remagnetization [11]. It is reported to occur with unambiguous (low-latitude) Cambrian directions in the Rome and Lamotte Formations and in the Colorado intrusives and Wichita granites (R.B. French, D.H. Alexander and R. Van der Voo, written communication, 1976; [13, p. 5636]).

A pole has been reported from the Late Cambrian Wilberns Formation of the Llano uplift, Texas [14,13]. Calculated from reversely magnetized rocks obtained from seven sites, the pole for the Wilberns is identical to within 1° of the pole reported here for normally magnetized Tapeats Sandstone. Without the poles for the Muav and Abrigo Formations, the pole for the Wilberns would be interpreted simply to reflect a Cambrian direction that predated any northward motion of the apparent polar wandering path. Such would be the case if high-latitude poles in Cambrian rocks are the consequence of a post-Cambrian remagnetization, although such remagnetization remains to be demonstrated for the Arizona rocks. The Arizona Devonian pole, which displays little apparent secondary magnetization and which in central Arizona has been superimposed locally on a bleached zone at the top of the Tapeats Sandstone, is considered to reflect the direction of the paleomagnetic field during Devonian time. In support of this Martin [15] has reported a high-latitude pole (45°N, 120°E) from reversely polarized mid-Devonian carbonate rocks in Ohio (Fig. 5), at the base of which are normally magnetized strata with an intermediate-latitude pole (24°N, 164°E). In the Ohio carbonates, the carrier of the remanence is apparently a titanomagnetite, whereas the carrier in the Arizona carbonates is hematite. Other poles from North America suggest a fairly regular progression of the polar wandering path during Ordovician to Devonian time. Intermediate-latitude poles (Fig. 5) have been reported from isotopically dated plutonic rocks of Cambrian to Ordovician age from central Colorado [10], from the Middle Ordovician Trenton Limestone [16], and from the Late Ordovician Beemerville Alkaline Complex of New Jersey [17]. These poles, which are summarized in Van der Voo et al. [13, table 2], and the high-latitude Devonian poles of Arizona and Ohio, indicate that large-scale apparent polar wandering occurred in an interval between Late Cambrian and Late Devonian time. The difference in latitude (7–10°) between the mid-Devonian Ohio pole and the pole for latest Middle(?) and early Late Devonian rocks of Arizona may, in part, reflect northward polar wandering at this time.

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